

Palletooru Smart Farmer Assistant

Blockchain & AI Powered AgriTech Platform

A role-based digital agriculture platform designed to eliminate middlemen, increase transparency, and empower farmers using Blockchain, Artificial Intelligence, and Machine Learning.

Key Features

- Farmer Module: Crop registration, QR generation after harvest, AI advisory, ML disease detection.
- Consumer Module: View harvested crops, scan QR codes, verify blockchain authenticity.
- Government Dashboard: District analytics, fertilizer usage trends, harvest forecasts, disease monitoring.

Authentication

- Role-based login (Farmer / Consumer / Government)
- Email & Password authentication
- Google OAuth integration

Blockchain Integration

- Crop data hashed after harvest completion
- Hash stored on Blockchain (Ethereum/Polygon Testnet)
- QR links to verified crop record
- Tamper-proof transparency system

AI & Machine Learning

- AI advisory using weather + crop stage
- Irrigation and fertilizer suggestions
- CNN/MobileNet-based disease detection

- Treatment and preventive recommendations

Real-Time Weather Integration

- Location-based weather data
- Temperature, humidity, rain probability
- Weather alerts

Tech Stack

- Frontend: React.js
- Backend: Node.js + Express
- Database: MongoDB / Firebase
- Blockchain: Ethereum / Polygon
- Machine Learning: TensorFlow / PyTorch